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Medical systems, devices, and related methods

Abstract

A medical device system includes an insertion device and a medical device. The insertion device includes an insertion device handle, including a port on a handle body. The insertion device also includes an insertion device shaft extending from the insertion device handle. The insertion device shaft includes a working channel connected to the port. The medical device includes a medical device handle, including a movable handle portion and a stationary handle portion. The movable handle portion includes a ball portion movably positioned within a cavity in the stationary handle portion. The medical device also includes a medical device shaft. The medical device shaft is configured to be delivered through the port in the insertion device handle and through the working channel in the insertion device shaft. Movement of the movable handle portion relative to the stationary handle portion controls movement of a distal portion of the medical device shaft.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This present disclosure claims priority to U.S. Provisional Patent Application No. 63/216,548, filed on Jun. 30, 2021, the disclosure of which is incorporated herewith by reference.

TECHNICAL FIELD

(1) Various aspects of this disclosure generally relate to medical systems, devices, and methods for manipulating or treating tissue or other material within a body. In particular, aspects of the disclosure relate to medical systems, devices, and methods for performing a medical procedure using, in some embodiments, a medical device capable of being inserted through an insertion device and into the body to treat a treatment site and deflecting or otherwise positioning the medical device relative to a distal end of the insertion device.

BACKGROUND

(2) A wide variety of medical techniques and instruments have been developed for diagnosis and/or treatment within a patient's body, such as within a patient's gastrointestinal (GI) tract. Endoscopic sub-mucosal dissection (ESD), endoscopic sub-mucosal resection (ESR), mucosal resection (EMR), polypectomy, mucosectomy, etc., are minimally invasive treatment methods for both malignant and non-malignant lesions. Endoscopic medical procedures, such as, for example, ESR, may be used to excise sessile adenomas or other unwanted tissue (e.g., tumors attached to a bodily surface) from the surface of an anatomical lumen (e.g., stomach, esophagus, colon, etc.). Such procedures often require the resection of one tissue plane while leaving an underlying tissue plane intact, or other precise treatments. Commonly, snares or other medical devices are used during such medical procedures, for resecting tissue from a treatment site. However, many conventional snares or medical devices operate in only one degree of freedom, and deflection of the snare (or another end effector) is often limited and/or dependent on the tip deflection of an endoscope or other device used for insertion into the patient. Furthermore, devices that may allow for separate deflection may require a number of operators, multiple hands, various movements of the hands, wrists, arms, etc. of the operator(s), an increased cognitive load on the operator(s) to obtain the desired movement, and/or other issues. These concerns may increase the duration, costs, and risks of the medical procedure. The devices and methods of this disclosure may rectify some of the deficiencies described above or address other aspects of the art.

SUMMARY

(3) Examples of this disclosure relate to, among other things, systems, devices, and methods for performing one or more medical procedures. Each of the examples disclosed herein may include one or more of the features described in connection with any of the other disclosed examples.

(4) In one example, a medical device system may include an insertion device and a medical device. The insertion device may include an insertion device handle, which may include a port on a handle body. The insertion device also may include an insertion device shaft extending from the insertion device handle. The insertion device shaft may include a working channel connected to the port. The medical device may include a medical device handle, which may include a movable handle portion and a stationary handle portion. The movable handle portion may include a ball portion movably positioned within a cavity in the stationary handle portion. The medical device also may include a medical device shaft. The medical device shaft may be configured to be delivered through the port in the insertion device handle and through the working channel in the insertion device shaft. Movement of the movable handle portion relative to the stationary handle portion may control movement of a distal portion of the medical device shaft.

(5) The medical device system may include one or more of the following features. The medical device system may further include a plurality of steering wires. Each of the plurality of steering wires may be coupled to the ball portion at a proximal end and to the distal portion of the medical device shaft at a distal end. The distal ends of the plurality of steering wires may be coupled to a ring at the distal portion of the medical device shaft. The plurality of steering wires may include

four steering wires coupled to the ball portion. The ball portion may include four wire mounts at locations 90 degrees apart from each other around a circumference of the ball portion. Each wire mount may include a crimping slot to couple each steering wire to the ball portion.

(6) The medical device may include an actuation wire and a movable body coupled to the actuation wire. The actuation wire may be movable relative to the medical device shaft by movement of the movable body relative to the stationary handle portion. The medical device may include a distal electrode, and movement of the movable body may manipulate the actuation wire to extend or retract the distal electrode relative to a distal end of the medical device shaft. The medical device handle may include a fluid port and/or a cautery hub.

(7) The medical device system may further include a cautery hub on the medical device handle and an actuation wire extending through at least a portion of the medical device handle and the medical device shaft. The actuation wire may be electrically connected to the cautery hub. The medical device system may further include an adapter positioned between the port on the handle of the insertion device and the medical device handle. The adapter may include an arced portion and/or may be at least partially flexible. The adapter may be removably coupled to the insertion device and may include one or more slits on a distal portion. The adapter may include a lock nut that is movable along the distal portion to control a width of each of the one or more slits.

(8) The medical device system may further include a guide wire device. The guide wire device may include a guide wire handle and a guide wire. The guide wire may include a main wire portion, a distal wire portion, and a movable pull wire coupled to the distal wire portion. The guide wire handle may include a casing, a roller rotatably coupled to the casing, and a button that is movable within a channel in the roller. The roller may be coupled to the pull wire to control extension and/or rotation of the pull wire. A portion of the pull wire may be coupled to the distal wire portion via a coupling. The coupling may extend over less than an entire outer circumference of the portion of the pull wire. The movable handle portion may be a joystick. The stationary handle body may be a main handle body positioned distal to the joystick.

(9) In another aspect, a medical system may include a medical device and a guide wire device. The medical device may include a medical device handle. The medical device handle may include a movable handle portion and a stationary handle portion. The movable handle portion may include a ball portion movably positioned within a cavity in the stationary handle portion. The medical device may also include a medical device shaft extending from the medical device handle. The medical device may also include a plurality of wires. The plurality of wires may be coupled to the ball portion of the movable handle portion and to a ring at a distal portion of the medical device shaft. Movement of the ball portion of the movable handle portion within the cavity in the stationary handle portion may manipulate a distal portion of the medical device shaft. The guide wire device may include a guide wire handle and a guide wire. The medical device shaft may include one or more openings or tubes along one or more outer portions of the medical device shaft configured to receive a portion of the guide wire.

(10) The medical system may include one or more of the following features. The guide wire device may include a main wire portion, a distal wire portion, and a movable pull wire coupled to the distal wire portion. The guide wire handle may include a casing, a roller rotatably coupled to the casing, and a button that is movable within a channel in the roller. The roller may be coupled to the pull wire to control extension and/or rotation of the pull wire. The medical system may further include an insertion device. The insertion device may include an insertion device handle having a port to receive a portion of the medical device shaft and a portion of the guide wire. The insertion device may also include an insertion device shaft extending from the insertion device handle. The insertion device shaft may include a lumen in communication with the port. The insertion device may also include a control device coupled to a portion of the insertion device handle. The control device may be movable to control a position of a distal end of the insertion device shaft.

(11) In yet another aspect, a medical device may include a medical device handle. The medical

device handle may include a joystick portion and a stationary handle portion. The joystick portion may be positioned distal to the stationary handle portion and may include a ball portion movably positioned within a cavity in the stationary handle portion. The medical device may also include a medical device shaft extending from the medical device handle. The medical device may include a plurality of wires. The plurality of wires may be coupled to the ball portion of the joystick portion and to a ring at a distal portion of the medical device shaft. Movement of the ball portion of the movable handle portion within the cavity in the stationary handle portion may manipulate a distal portion of the medical device shaft.

(12) The medical device may include one or more of the following features. The medical device may include a movable body coupled to the stationary handle portion. The medical device may further include an actuation wire extending from the movable body to an end effector at a distal end of the medical device shaft. Movement of the movable body may control an extension or retraction of the end effector.

(13) It may be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the disclosure, as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate exemplary aspects of the disclosure and together with the description, serve to explain the principles of the disclosure.

(2) FIG. 1 illustrates a perspective view of an exemplary medical system, including an insertion device, a medical device, and an adapter, according to aspects of this disclosure.

(3) FIGS. 2A-2C illustrate various views of the adapter, according to aspects of this disclosure.

(4) FIGS. 3A-3D illustrate various views of various portions of the medical device, including a medical device handle and a medical device shaft, according to aspects of this disclosure.

(5) FIGS. 4A and 4B illustrate various cutaway views of the medical device shaft, according to aspects of this disclosure.

(6) FIG. 5 illustrates the medical system with the medical device handle and a distal portion of the medical device shaft in various configurations, according to aspects of this disclosure.

(7) FIGS. 6A and 6B illustrate another medical device system including an insertion device, a medical device, and a guide wire device, where the medical device includes a medical device handle, a medical device shaft, and a distal end, according to aspects of this disclosure.

(8) FIG. 7 illustrates the medical device, with the medical device handle and the medical device shaft in various configurations, according to aspects of this disclosure.

(9) FIGS. 8A and 8B illustrate various aspects of the medical device handle and medical device shaft, according to aspects of this disclosure.

(10) FIG. 9 illustrates the medical device handle and the medical device shaft in various configurations, according to aspects of this disclosure.

(11) FIGS. 10A-10D illustrate various views of the medical device shaft, including cross-sections along various portions of the medical device shaft, according to aspects of this disclosure.

(12) FIG. 11A illustrates a perspective view of the guide wire device of FIG. 7, and FIG. 11B illustrates a cross-sectional view of the guide wire device of FIG. 7, according to aspects of this disclosure.

(13) FIGS. 12A-12C illustrate portions of the guide wire device, including a handle portion and a distal end of a guide wire shaft, according to aspects of this disclosure.

(14) FIG. 13 illustrates another medical device system including an insertion device, a medical device, and a guide wire device, according to aspects of this disclosure.

DETAILED DESCRIPTION

(15) The terms “proximal” and “distal” are used herein to refer to the relative positions of the components of an exemplary medical system and exemplary medical devices. When used herein, “proximal” refers to a position relatively closer to the exterior of the body or closer to a medical professional using the medical system or medical device. In contrast, “distal” refers to a position relatively further away from the medical professional using the medical system or medical device, or closer to the interior of the body. As used herein, the terms “comprises,” “comprising,” “having,” “including,” or other variations thereof, are intended to cover a non-exclusive inclusion, such that a system, device, or method that comprises a list of elements does not include only those elements, but may include other elements not expressly listed or inherent thereto. Unless stated otherwise, the term “exemplary” is used in the sense of “example” rather than “ideal.” As used herein, the terms “about,” “substantially,” and “approximately,” indicate a range of values within $\pm 10\%$ of a stated value.

(16) Examples of this disclosure include systems, devices, and methods for facilitating and/or improving the efficacy, efficiency, and/or safety of a medical procedure. Embodiments of the disclosure may relate to systems, devices, and methods for performing various medical procedures and/or treating portions of the large intestine (colon), small intestine, cecum, esophagus, stomach, any other portion of the gastrointestinal tract, lungs, and/or any other suitable patient anatomy. Various embodiments described herein include single-use or disposable medical devices. Some aspects of the disclosure may be used in performing an endoscopic, arthroscopic, bronchoscopic, ureteroscopic, colonoscopic, or other type of procedure. For example, the disclosed aspects may be used with duodenoscopes, bronchoscopes, ureteroscopes, colonoscopes, catheters, diagnostic or therapeutic tools or devices, or other types of medical devices. One or more of the elements discussed herein could be metallic, plastic, or include a shape memory metal (such as nitinol), a shape memory polymer, a polymer, or any combination of biocompatible materials.

(17) Reference will now be made in detail to examples of the disclosure described above and illustrated in the accompanying drawings. Wherever possible, the same reference numbers will be used throughout the drawings to refer to the same or like parts. It is noted that one or more aspects of the medical systems or devices discussed herein may be combined and/or used with one or more aspects of other medical systems or devices discussed herein.

(18) FIG. 1 illustrates a perspective view of an exemplary medical system **10** that includes an insertion device **12**, a medical device **14**, and an adapter **16**. Insertion device **12** includes an insertion device handle **18**, including a handle body, and an insertion device shaft **20**, for example, extending from a distal end of handle **18** to a distal end **20A**. Insertion device **12** also includes at least one working channel (not shown), for example, extending from insertion device handle **18** to a distal opening **20B** in distal end **20A**. Insertion device **12** also includes a port **22**, for example, in handle **18**, and port **22** may connect to the working channel. Insertion device **12** may include a control device **24**, for example, on a portion of handle **18**, and one or more portions of control device **24** may be manipulated (e.g., rotated) to control a deflection of a portion of medical device shaft **20**, for example, distal end **20A**.

(19) Medical device **14** may include a medical device handle **26** and a medical device shaft **28** extending from medical device handle **26** to a distal end **30**. Medical device handle **26** may include a movable handle portion or a main handle body **26A** and a stationary handle portion or a joint (e.g., a ball joint), referred to herein as socket body **26B**. Main handle body **26A** is movable relative to and positioned proximal to socket body **26B**. As discussed in detail below, movement of main handle body **26A** relative to socket body **26B** may control the movement of one or more steering wires (FIGS. 3A, 3D, and 3E) and control a deflection of distal end **30** of medical device shaft **28**. Additionally, medical device shaft **28** may be delivered through port **22**, for example, via adapter **16**, and through the working channel to extend from distal end **20A** of shaft **20** of insertion device **12**. In these aspects, distal end **30** of medical device shaft **28** may be extended from distal end **20A**

of insertion device **12**, and may be controlled via one or more of control device **24**, movement of medical device handle **26** relative to insertion device handle **18**, and movement of main handle body **26A** relative to socket body **26B**.

(20) Insertion device **12** may be a duodenoscope, an endoscope, a colonoscope, an ureteroscope, a bronchoscope, etc., or any other like device having a handle and a shaft. As mentioned, insertion device **12** may include control device **24**, for example, on a proximal portion of handle **18**. Control device **24** may be movable (e.g., rotatable) relative to handle **12**, and may control the movement of a portion (e.g., distal end **20A**) of insertion device shaft **20**. Control device **24** may include one or more dials or knobs. As shown in FIG. **1**, control device **24** may include first and second knobs **24A** and **24B**, for example, each rotatable to deflect distal end **20A** along two different planes. Control device **24** may include one or more locking mechanisms. As shown in FIG. **1**, control device **24** may include two locking mechanisms **24C** and **24D**, for example, each engageable with one of knobs **24A** and **24B** to lock and/or unlock the position of the knob, and thus lock and/or unlock the position of distal end **20A** of insertion device shaft **20**. Alternatively, control device **24** may actuate or move one or more elevators in insertion device shaft **20**, or otherwise actuate a cable driven function of insertion device **12**.

(21) Insertion device **12** may include a conduit **32**. For example, insertion device handle **18** may be coupled to conduit **32**. Conduit **32** may connect insertion device handle **18** to an external power source, processing software, one or more displays, one or more memory or storage devices, etc., for example, via an umbilicus (not shown). In this aspect, insertion device **12** may include one or more illumination devices and/or cameras at distal end **20A**, which may be powered and/or connected to processing software, one or more displays, a memory, etc. via one or more communication wires (not shown) within insertion device **12** and via conduit **32**. Additionally, conduit **32** may connect insertion device handle **18** to one or more fluid sources, for example, an air source, a water source, etc. Conduit **32** may also connect insertion device handle **18** to a suction source. In these aspects, one or more valves coupled to or received within one or more apertures (not shown) in insertion device handle **18** may control the delivery of air or water and/or the application of suction through insertion device **12** to the area distal to distal end **20A** of insertion device shaft **20**.

(22) As mentioned, medical device **14** includes medical device handle **26** and medical device shaft **28** extending from medical device handle **26** to distal end **30**. Medical device handle **26** includes main handle body **26A** and joint (e.g., a ball joint), referred to herein as socket body **26B**, with main handle body **26A** being movable relative to socket body **26B**. For example, socket body **26B** is shown as being partially transparent in FIG. **1**. As shown, a distal portion or a ball portion **34** of main handle body **26A** may be positioned within socket body **26B**. As discussed below, one or more control elements (e.g., steering wires) may be connected to ball portion **34**, and the one or more control elements may be coupled to a distal portion of medical device shaft **28**. Movement of main handle body **26A** relative to socket body **26B** may control the movement of the one or more control elements, and also control the deflection of the distal portion (i.e., distal end **30**) of medical device shaft **28**. Additionally, although not shown, medical device handle **26** may include one or more frictional and/or locking elements to help control and/or lock the relative movement of main handle body **26A** relative to socket body **26B**.

(23) Medical device handle **26** also may include a port **36A** configured to receive fluid, and a hub **36B** configured to receive electrical energy similar to an electrical plug or socket. Port **36A** and hub **36B** may be positioned on proximal portions of medical device handle **26**. Distal end **30** of medical device shaft **28** may include an end effector, for example, an energy delivery portion or an electrode portion **38** (hereinafter “electrode **38**”). Electrode **38** is electrically connected to hub **36B**, and may include a channel fluidly connected to, or otherwise in fluid communication with, port **36A**. Additionally, as shown in FIG. **1**, electrode **38** may include a distal tip **38A** and an electrode shaft **38B**. In some aspects, distal tip **38A** may be wider (e.g., in a lateral direction away from a longitudinal direction of electrode shaft **38B**) than electrode shaft **38B**. For example, distal tip **38A**

may include a mushroom-like or semi-spherical tip. In some aspects, the size and/or shape of distal tip **38A** may help the user to deliver energy and/or treat tissue. Furthermore, although not shown, distal end **30** of medical device shaft **28** may include an end cap surrounding a portion of electrode **38**, for example, to help insulate medical device shaft **28** from electrode **38**.

(24) Medical device handle **26** may include a movable body **40**, for example, on a portion of main handle body **26A**. Movable body **40** may be a knob or other mechanism movable (e.g., slideable) within a track formed in main handle body **26A**, and movable body **40** may control the position of electrode **38**, for example, relative to distal end **30** of medical device shaft **28**. In this aspect, and as discussed below, medical device **14** may include a pull wire, actuation wire, a drive wire, or other connection(s) between movable body **40** and electrode **38**, for example, coupled to electrode shaft **38B**. In this example, movement of movable body **40** relative to main handle body **26A** in a first direction (e.g., the distal direction) may extend electrode **38** relative to medical device shaft **28** (e.g., move electrode **38** distally relative to a distal end **30** of medical device shaft **28**). Similarly, movement of movable body **40** relative to main handle body **26A** in a second direction (e.g., the proximal direction) may retract electrode **38** relative to medical device shaft **28** (e.g., move electrode **38** proximally relative to a distal end **30** of medical device shaft **28**).

(25) In some aspects, main handle body **26A** may be formed of two halves, for example, that form the track in which movable body **40** is movable. For example, although not shown, one half of main handle body **26A** may include unthreaded holes, and another half of main handle body **26A** may include threaded holes. In this aspect, the halves of main handle body **26A** may be coupled via one or more screws. Alternatively, the halves of main handle body **26A** may be coupled via an adhesive, a snap-fit, or other appropriate coupling mechanism.

(26) In some aspects, medical device handle **26** may be coupled to a fluid source via port **36A**. Port **36A** may be in fluid communication with electrode **38** via an internal lumen (not shown), which may extend through medical device handle **26** and medical device shaft **28**. In these aspects, electrode **38** may include an internal lumen with one or more outlets **38C**, for example, in a distal portion of electrode **38**, to deliver fluid to a treatment site, for example, to inject fluid below a layer of tissue (e.g., the mucosal layer) and help lift the layer of tissue. Port **36A** may include a one-way valve, a luer, a seal, threading, and/or any appropriate connection or mating element to help maintain a secure connection between medical device handle **26** and the fluid source, minimize or prevent back-flow (e.g., fluid flowing proximally out of port **36A**), and/or inhibit, minimize, or prevent leakage. In at least one example, port **36A** may include a one-way valve having an outer housing containing an inner elastomeric and/or gelatinous sealing member. Although not shown, one or more lumens, tubes, conduits, channels, etc. may be positioned within insertion device **12** and/or medical device **14** and may fluidly connect port **36A** to electrode **38**, for example, to deliver fluid out of electrode outlet **38C**. In some examples, medical device handle **26** does not include a port **36A**, for example, when medical device **14** is not used for fluid delivery.

(27) In some aspects, medical device handle **26** may be coupled to an energy source via hub **36B**. Hub **36B** may include one or more prongs or pins to couple to the energy source. Hub **36B** may be electrically coupled to electrode **38** via one or more conductive elements, which may be electrically coupled to the one or more prongs or pins of hub **36B** and extend through medical device handle **26** and through at least a portion of medical device shaft **28**. The energy source may be, for example, an electrocautery source, a radio frequency generator, a heating source, a current generator, etc. In other aspects, the energy source may be a part of medical device handle **26** (e.g., an internal battery in medical device handle **26**). Although not shown, one or more actuators (e.g., foot pedals, buttons, switches, etc.) may control the delivery of energy from the energy source, and thus control the delivery of energy from electrode **38**. Alternatively, medical device **14** may be used to apply suction. In this example, medical device handle **26** may include a suction port, for example, to couple medical device handle **26** and medical device shaft **28** to a suction source.

(28) In at least one aspect, medical device **14** may be used for monopolar electrosurgery, and may

include a return electrode positioned remotely from electrode **38** on or otherwise adjacent to the subject. In other aspects, medical device **14** may be used for bipolar electrosurgery. In such instances, electrode **38** may include an active electrode portion, and a return electrode may be provided at or near another portion of electrode **38**, medical device shaft **28**, and/or insertion device shaft **20**. In at least one example, two conductive elements may run through medical device shaft **28**, where the conductive elements may be electrically isolated from each other, allowing one conductive element to conduct energy to the active electrode and the other conductive element to conduct energy from a return electrode.

(29) Referring to FIGS. **1** and **2A-2C**, adapter **16** may be a substantially tubular element, and may help couple medical device **14** to insertion device **12**. Adapter **16** includes a proximal portion **16A**, for example, configured to receive a portion of medical device shaft **28**, and about a distal portion of medical device handle **26**, for example, a distal portion of socket body **26B**. Adapter **16** also includes a distal portion **16B**, for example, configured to be coupled to a portion of port **22**. As mentioned, adapter **16** may be positioned on or over or otherwise coupled to port **22** of insertion device handle **18**, and medical device shaft **28** may be inserted through adapter **16**, into port **22**, and through the working channel of insertion device **12** to distal end **20A** of insertion device shaft **20**. Port **22** may include a one-way valve, a luer, a seal, threading, and/or any appropriate connection or mating element to help maintain a secure connection between insertion device handle **18** and adapter **16**, inhibit, minimize, or prevent back-flow (e.g., fluid flowing proximally out of port **22**), and/or inhibit, minimize, or prevent leakage.

(30) Additionally, adapter **16** may help to form a fulcrum or pivot point for medical device handle **26**, for example, such that main handle body **26A** may move relative to socket body **26B**. Adapter **16** may also help to allow for medical device **14**, and thus medical device shaft **28** to move proximally or distally relative to insertion device **12**, allowing for distal end **30** of medical device shaft **28** to move relative to insertion device distal end **20A**.

(31) Furthermore, distal portion **16B** of adapter **16** may include one or more slits **42A**, for example, extending longitudinally in a portion of distal portion **16B**. In some configurations, a distal portion of the one or more slits **42A** may be wider than a proximal portion of the one or more slits **42A**. Adapter **16** may also include a lock nut **42B** and threading **42C**, for example, on distal portion **16B**. As shown in FIG. **2A**, lock nut **42B** may be movable (e.g., rotatable) on threading **42C**. Although not shown, lock nut **42B** may include an internal threading that is configured to interact with threading **42C**. Lock nut **42B** may include one or more protrusions or grip features **42D**, for example, to help the user manipulate lock nut **42B**.

(32) Rotation of lock nut **42B** on threading **42C** in a first direction (e.g., clockwise) may translate lock nut **42B** along adapter **16** in a first direction (e.g., distally), and rotation of lock nut **42B** on threading **42C** in a second direction (e.g., counterclockwise) may translate lock nut **42B** along adapter **16** in a second direction (e.g., proximally). In a distalmost position, lock nut **42B** may at least partially overlap with one or more slits **42A**. For example, threading **42C** may be adjacent to or partially overlap with one or more slits **42A**.

(33) Additionally, as shown in FIGS. **2B** and **2C**, distal portion **16B** may also include one or more projections **42E**, for example, extending radially inward. As shown in FIG. **2C**, port **22** may include one or more extensions **22A**, for example, extending radially outward. In this aspect, when adapter **16** is coupled to port **22**, projections **42E** and extensions **22A** may interact to help secure adapter **16** to port **22**. For example, with lock nut **42B** proximally retracted, distal portion **16B** may be in a first configuration, for example, with one or more slits **42A** in open configuration(s). Adapter **16** may be positioned over a portion of port **22** of insertion device **12** with distal portion **16B** in the first configuration. Then, lock nut **42B** may be distally translated, for example, via rotation and interaction with threading **42C**. The movement of lock nut **42B** may tighten distal portion **16B**, for example, by reducing the size or width of slits **42A**, and transitioning distal portion **16B** into a second configuration, for example, with one or more slits **42A** in closed configuration(s). As shown

in FIG. 2C, with slits 42A in closed configurations, portions of projection(s) 42E may at least partially abut portions of extension(s) 22A. In the closed configurations, the interaction of projection(s) 42E and extension(s) 22A may, help maintain a secure connection between port 22 and adapter 16, inhibit, minimize, or prevent back-flow (e.g., fluid flowing proximally out of port 22 and/or adapter 16), and/or inhibit, minimize, or prevent leakage.

(34) Adapter 16 may include a bent or arced portion 16C, for example, in a middle portion of adapter 16 between proximal portion 16A and distal portion 16B. Arced portion 16C may help the user to maintain a comfortable, or otherwise suitable, position while handling various components of medical system 10 (e.g., insertion device 12 and medical device 14). In some aspects, adapter 16 may be at least partially flexible, for example, allowing a user to manipulate (e.g., bend) adapter 16 and thus position medical device 14 in a position that is comfortable or otherwise suitable for the user and the medical procedure. Furthermore, although not shown, adapter 16 may include a one-way valve, a luer, a seal, and/or any appropriate connection or mating element to help maintain a secure connection between port 22 and adapter 16 and/or around medical device shaft 28, inhibit, minimize, or prevent back-flow (e.g., fluid flowing proximally out of adapter 16), and/or inhibit, minimize, or prevent leakage.

(35) FIGS. 3A-3D illustrate various features of the medical device 14, including main handle body 26A and socket body 26B of medical device handle 26. FIG. 3A illustrates socket body 26B as transparent, allowing internal features of medical device handle 26 to be visible. As shown, ball portion 34 of main handle body 26A may be positioned within a cavity 44 within socket body 26B. As shown in FIGS. 3A and 3D, a proximal portion of cavity 44 may include one or more bulges or ledges 44A, for example, to help retain ball portion 34 within cavity 44. Furthermore, as shown in FIGS. 3A and 3D, socket body 26B may include one or more protrusions 44B, for example, on an inward face of socket body 26B to help form cavity 44. Although not shown, a proximal end of cavity 44 may include one or more gaskets or other sealing elements to help retain ball portion 34 within socket body 26B. Moreover, a portion of main handle body 26A proximal to ball portion 34 may include a widened portion 46, which may help to limit the relative movement of main handle body 26A relative to socket body 26B. As discussed below, ball portion 34 may include one or more wire mounts 48. For example, ball portion 34 may include four wire mounts 48 positioned approximately 90 degrees from each other around a circumference of ball portion 34.

(36) In these aspects, a plurality of steering wires 50 may be coupled to ball portion 34 via wire mounts 48. Movement of ball portion 34, for example, via movement of main handle body 26A relative to socket body 26B, may retract one or more of steering wires 50, for example, to deflect a portion of medical device shaft 28. Additionally, an actuation wire 52 may extend through ball portion 34, socket body 26B, and medical device shaft 28 and to electrode 38, for example, to control the extension or retraction of electrode 38.

(37) Socket body 26B includes cavity 44 to receive a portion of ball portion 34. Socket body 26B may be formed of two halves that are coupled around ball portion 34, for example, via one or more screws, bolts, etc. positioned within coupling holes 44C. In one aspect, coupling holes 44C in one half of socket body 26B may include unthreaded holes, and one half of socket body 26B may include threaded holes. Additionally, socket body 26B includes a cylindrical portion 54. For example, each half of socket body 26B may include a cylindrical portion 54 (i.e., a semi-cylindrical portion), and when the halves of socket body 26B are coupled together, the halves form cylindrical portion 54. Although referred to as a cylindrical portion herein, it is contemplated that portion 54 may have any suitable cross-sectional shape and may be, for example, tapered.

(38) Although not shown, in one or more aspects, main handle body 26A may include a cavity, and socket body 26B may include a ball portion. In this aspect, the ball portion of socket body 26B may be positioned within the cavity of main handle body 26A. In this aspect, main handle body 26A may include one or more wire mounts to couple one or more steering wires to main handle body 26A. Accordingly, movement of main handle body 26A relative to socket body 26B moves (e.g.,

retracts) the one or more steering wires relative to socket body **26B** to deflect a portion of medical device shaft **28**.

(39) Cylindrical portion **54** also includes a channel **54A** to receive a portion of medical device shaft **28**. Additionally, channel **54A** may include a widened channel portion or slot **54B**. In some aspects, channel **54A** may be generally cylindrical, and slot **54B** may be generally rectangular. A crimp **54C** may be positioned around a portion of medical device shaft **28** within slot **54B**, for example, to help couple medical device shaft **28** to medical device handle **26**. In this aspect, a portion of medical device shaft **28** and crimp **54C** may be positioned within channel **54A** and slot **54B** of one half of socket body **26B**, and then the other half of socket body **26B** may be coupled to the first half of socket body **26B** to enclose the portion of medical device shaft **28** and crimp **54C**. Crimp **54C** may move (i.e., proximally and/or distally) within slot **54B** as insertion device shaft **20**, and thus medical device shaft **28**, moves (e.g., bends) within the patient, for example, as insertion device shaft **20** and medical device shaft **28** traverse a tortuous path within the patient. Alternatively, a portion of medical device shaft **28** may be coupled to a portion of medical device handle **26** via heat shrinking or other coupling mechanism.

(40) Medical device handle **18** may also include a handle cap **54D**. Handle cap **54D** may be coupled to a distal portion of cylindrical portion **54**. In some aspects, the distal portion of cylindrical portion **54** may include a threading, and handle cap **54D** may include an internal threading, such that handle cap **54D** may be screwed onto the distal portion of cylindrical portion **54**. Handle cap **54D** may also include an internal lumen, which allows medical device shaft **28** to extend from socket body **26B** of medical device handle **26**. Handle cap **54D** may help to couple portions of socket body **26B** (e.g., cylindrical portions of two halves). Handle cap **54D** may also help to couple medical device shaft **28** to medical device handle **26**. In one or more aspects, handle cap **54D** may include one or more protrusions or grip features **54E**, for example, to help the user manipulate handle cap **54D**.

(41) FIG. 3B illustrates a perspective view of ball portion **34** of main handle body **26A**, including wire mounts **48**. Each wire mount **48** may include a wire slot **48A**, for example, extending in a longitudinal direction of main handle body **26A**. Wire slots **48A** may help align each steering wire **50** with each wire mount **48**, which may help prevent steering wire **50** from crimping, etc. Each wire slot **48A** may be connected to or otherwise adjacent to a wire opening **48B**, for example, extending approximately perpendicular to wire slot **48A** along an outer circumference of ball portion **22**. For example, wire openings **48B** may be radially inward extending indentations in the outer circumference of ball portion **22**. Additionally, wire openings **48B** may include crimping slots **48C**. Crimping slots **48C** may be formed by ball portion **34**, for example, by a base portion **48D** and an extended portion **48E**. In these aspects, one steering wire **50** may be positioned within wire slot **48A**, wire opening **48B**, and crimping slot **48C**. Extended portion **48E** may be brought toward (e.g., compressed toward) base portion **48D**, for example, with pliers, a vice, or other force applying mechanism, in order to form a crimp and secure steering wire **50** within wire mount **48**, for example, within wire slot **48A**. This process may be performed for each steering wire **50** and each wire mount **48**. Additionally, any portion of steering wire **50** that extends proximal to or beyond wire mount **48** may be cut, tied off, or otherwise removed or positioned to not interfere with the movement of main handle body **26A**. Furthermore, the crimp that secures each steering wire **50** may be positioned approximately perpendicular to the longitudinal axis of main handle body **26A**. Moreover, the crimp that secures each steering wire **50** may be substantially cylindrical and/or free to rotate. In this aspect, as main handle body **26A** and ball portion **34** move, the crimp may pivot about its own axis, which may be a different axis than main handle body **26A** and ball portion **34**. The above-discussed arrangement of the crimp securing steering wire **50** relative to wire mount **48** and ball portion **34** of main handle body **26A** may help to reduce, limit, or otherwise mitigate risks of steering wire **50** kinking or otherwise deforming as handle body **26A** and ball portion **34** move.

(42) FIG. 3C is a cross-sectional view of ball portion 34 and a portion of main handle body 26A. As shown, ball portion 34 includes a generally spherical outer surface, which helps allow for ball portion 34 to move within socket body 26B. FIG. 3C also illustrates wire mounts 48, each including wire slot 48A. Furthermore, ball portion 34 includes a central opening 34A, for example, to accommodate a conductive element, for example, actuation wire 52. Central opening 34A may widen from a proximal portion to a distal portion, for example, in a triangular or conical shape. In this aspect, actuation wire 52 may extend through central opening 34A, and actuation wire 52 may not contact ball portion 34 regardless of the position of main handle body 26A relative to socket body 26B. Moreover, ball portion 34 may include one or more coupling holes 34B, for example, to couple two halves of main handle body 26A.

(43) FIG. 3D illustrates a portion of medical device handle 26 in a first deflected position. As shown, main handle body 26A is positioned at a non-parallel angle relative to socket body 26B. For example, main handle body 26A is pivoted upward relative to socket body 26B. In this aspect, one control wire, for example, steering wire 50A, which is coupled to a bottom portion of ball portion 34 (as shown), is retracted proximally. As discussed below, proximal retraction of one of steering wires 50 via the movement of main handle body 26A relative to socket body 26B deflects a distal portion of medical device shaft 28. Nevertheless, because actuation wire 52 extends from a central portion of main handle body 26A, the movement of main handle body 26A relative to socket body 26B does not affect the position or movement of actuation wire 52. Instead, the movement of movable body 40 controls the position and movement of actuation wire 52.

(44) Furthermore, as mentioned above, socket body 26B includes ledge 44A to help retain ball portion 34 within socket body 26B. For example, ledge 44A may extend inward around a proximal end of socket body 26B, adjacent to cavity 44. Portions of socket body 26B may be coupled via screws, pins, or other coupling mechanisms being positioned within coupling holes 44C.

(45) It is noted that, although four steering wires 50 and four wire mounts 48 are discussed herein, this disclosure is not so limited. For example, medical device 14 may include two steering wires 50 and two wire mounts 48, three steering wires 50 and three wire mounts 48, five steering wires 50 and five wire mounts 48, etc. Furthermore, wire mounts 48 may be evenly positioned around a circumference of ball portion 34, or, although not shown, wire mounts 48 may be unevenly positioned around the circumference of ball portion 34 or may be located more proximal or more distal on ball portion 34.

(46) FIGS. 4A and 4B illustrate various details of medical device shaft 28. For example, FIG. 4A illustrates a lateral cross-section of medical device shaft 28 in a main portion 28A of medical device shaft 28, and FIG. 4B illustrates various portions of a distal portion 28B of medical device shaft 28.

(47) As shown in FIG. 4A, main portion 28A of medical device shaft 28 may be formed of an interior multi-lumen element 56, which may be formed of polytetrafluoroethylene ("PTFE") or other appropriate material, for example, via extrusion or other appropriate formation process. Steering wires 50 may be movably positioned within respective lumens 56A in multi-lumen element 56, for example, spaced approximately 90 degrees apart in the cross-section of multi-lumen element 56. Additionally, multi-lumen element 56 may include a central lumen 56B, for example, centrally located in multi-lumen element 56, and actuation wire 52 may be moveably positioned within central lumen 56B. It is noted that, if medical device 14 includes a different number of steering wires, then multi-lumen element 56 may include a different number and/or different spacing of lumens 56A such that each steering wire 50 is housed within a separate lumen of multi-lumen element 56. Additionally, main portion 28A of medical device shaft 28 may include one or more outer sheaths 58A, for example, formed of a polyether block amide ("PEBA") or other appropriate material. Outer sheath(s) 58A may be formed, for example, of PEBA may include a durometer between approximately 60 to 80, for example, approximately 72. Additionally, medical device shaft 28 may include one or more layers of braiding, for example, positioned between multi-

lumen element **56** and the one or more outer sheaths **58A**. Main portion **28A** may span a majority of the length of medical device shaft **28**, for example, approximately 70%, approximately 75%, approximately 80%, approximately 85%, approximately 90%, approximately 95%, or more of the length of medical device shaft **28**.

(48) Moreover, as shown in FIG. **4B**, medical device shaft **28** may include a distal portion **28B** with a different arrangement and/or different properties than that shown in FIG. **4A**. For example, distal portion **28B** may include lumens for steering wires **50** and actuation wire **52**, but may include a multi-lumen element (not shown) with more flexibility, for example, formed of PEBA, for example, with a durometer between approximately 25 to 55, for example between approximately 33 to 50. Additionally, distal portion **28B** may include an outer sheath **58B**, for example, formed of PEBA, with a durometer between approximately 20 to 50, for example, approximately 33. Distal portion **28B** may also include one or more layers of braiding **58C**, for example, positioned between the multi-lumen element and the one or more outer sheaths **58B**. In these aspects, distal portion **28B** of medical device shaft **28** may be more flexible than main portion **28A**, for example, for distal portion **28B** to be deflected by movement of steering wires **50**.

(49) FIG. **4B** also illustrates various features that may be incorporated at a distal portion **28B** of medical device shaft **28** in order to steer or deflect distal portion **28B**. For example, distal ends of each of steering wires **50** may be welded, adhered, or otherwise fixedly coupled to a ring **60**. Ring **60** may be substantially cylindrical and positioned radially outside of or around steering wires **50**. Ring **60** may be formed of steel or another appropriate metal or other material. Additionally, a distal portion **52A** of actuation wire **52** may be connected (directly or indirectly) to electrode **38** (FIG. **1**) to control the extension or retraction of electrode **38**. For example, distal portion **52A** may include a widened extension portion. Additionally, as discussed above, energy delivered through hub **36B** on medical device handle **26** may be delivered to electrode **38** via actuation wire **52**.

(50) FIG. **5** illustrates various configurations of system **10** and the different positions of medical device shaft **28**. As shown, medical device **14** is coupled to handle **18** of insertion device **12** via adapter **16**. Additionally, medical device shaft **28** is extended through insertion device shaft **20**. Main handle body **26A** may be pivoted in any direction relative to socket body **26B**, indicated by arrows A, B, C, and D. For example, the user may hold insertion device handle **18** with one hand, and may hold medical device handle **26** in another hand. The user may then move his or her wrist of the another hand in order to control main handle body **26A**. In this aspect, main portion **28A** of medical device shaft **28** may remain stationary, but distal end portion **28B** of medical device shaft **28** may be deflected in any direction, indicated by arrows E. For example, moving the user's wrist left or right will deflect distal end **30** of medical device shaft **28** right or left. Similarly, moving the user's wrist up or down, or side to side, will deflect distal end **30** of medical device shaft **28** down or up, or side to side.

(51) In one or more aspects, a right-side steering wire **50** may be coupled to a right side of ball portion **34** of main handle body **26A**. The right-side steering wire **50** may pass through the right-side lumen of multi-lumen element **56**, and the right-side steering wire **50** may then be coupled to a right-side portion of distal end **20A** of insertion device shaft **20**. A left-side steering wire **50** may be coupled to a left side of ball portion **34** of main handle body **26A**. The left-side steering wire **50** may pass through the left-side lumen of multi-lumen element **56**, and the left-side steering wire **50** may then be coupled to a left-side portion of distal end **20A** of insertion device shaft **20**.

Accordingly, moving main handle body **26A** to the right may cause the left-side steering wire **50** to be under tension, causing distal end **20A** of insertion device shaft **20** to bend or deflect to the left. Similarly, moving main handle body **26A** to the left may cause the right-side steering wire **50** to be under tension, causing distal end **20A** of insertion device shaft **20** to bend or deflect to the right. In another aspect, the right-side steering wire **50** may be coupled to the right side of ball portion **34** of main handle body **26A**, and the left-side steering wire **50** may be coupled to the left side of ball portion **34** of main handle body **26A**. The right-side steering wire **50** may pass through the left-side

lumen of multi-lumen element **56**, and the right-side steering wire **50** may then be coupled to a left-side portion of distal end **20A** of insertion device shaft **20**. Likewise, the left-side steering wire **50** may pass through the right-side lumen of multi-lumen element **56**, and the left-side steering wire **50** may then be coupled to a right-side portion of distal end **20A** of insertion device shaft **20**. In this aspect, moving main handle body **26A** to the right may cause the right-side steering wire **50** to be under tension, causing distal end **20A** of insertion device shaft **20** to bend or deflect to the right. Similarly, moving main handle body **26A** to the left may cause the left-side steering wire **50** to be under tension, causing distal end **20A** of insertion device shaft **20** to bend or deflect to the left.

(52) Insertion device shaft **20** may also be deflected, for example, via knobs **24A**, **24B**.

Furthermore, although not shown, medical device handle **26** may be retracted proximally relative to adapter **16** and handle **18** of insertion device **12** to retract medical device shaft **28** relative to insertion device shaft **20**, for example, to retract distal end **30** of medical device shaft **28**.

Additionally, the user may extend or retract electrode **38** from distal end **30** of medical device shaft **28** via movable body **40**, for example, via action from the user's thumb.

(53) FIGS. **6A** and **6B** illustrate another exemplary medical system **110**. As shown, medical system **110** includes an insertion device **112**, for example, similar to insertion device **12**. Medical system **110** also includes a medical device **114**. In some aspects, medical system **110** does not include an adapter. Furthermore, medical system **110** may include a guide wire device **162**. Medical device **114** may be coupled to and/or delivered to a treatment site through a handle **118** and shaft **120** of insertion device **112**, as discussed above. Guide wire device **162** may be coupled to medical device **112**, for example, via a slot or opening in a portion of medical device shaft **128** or medical device handle **126**.

(54) Medical device **114** includes a medical device handle **126** and a medical device shaft **128**. Medical device handle **126** may include a stationary handle portion or a handle body **126A**, a movable handle portion or a joystick **164** coupled to handle body **126A** via a ball-socket coupling, and a movable body **140**. Additionally, medical device handle **126** may include a handle extension **166**, for example, to be gripped by a user. Handle extension **166** may include a hub **136B**, for example, to couple an energy source to medical device **114**. Although not shown, medical device **114** may include one or more ports, for example, to couple a fluid source to medical device **114**. Medical device shaft **128** may be delivered through a port **122** in a handle **118** of insertion device **112**, as discussed above, and may be moveable proximally and distally to control the location of a distal end **130** of medical device shaft **128**. Moreover, joystick **164** may be movable to deflect distal end **130** of medical device shaft **128**, and movable body **140** may be movable (e.g., via a user's finger) to control the position of the portion of distal end **130** of medical device shaft **128**. In these aspects, medical device **114** may be a sphincterotome or other appropriate medical device.

(55) As shown in FIGS. **6A** and **6B**, distal end **130** of medical device shaft **128** may include a cautery wire **168**, for example, extending between a first portion of medical device shaft **128** and a second portion of medical device shaft **128**. For example, movement of movable body **140** may distally extend or proximally retract cautery wire **168**, for example, to control the shape or position of distal end **130** of medical device shaft **128**. Additionally, a guide wire **170** may extend through medical device shaft **128**. Guide wire device **162** may control the position of guide wire **170**, and thus also, in some aspects, may control the position of distal end **130** of medical device shaft **128**.

(56) FIG. **7** illustrates the movement of movable body **140** and the movement of cautery wire **168** in order to control the shape or position of distal end **130** of medical device shaft **128**. It is noted that medical device **114** is shown in FIG. **7** uncoupled from insertion device **112**, but that the movements disclosed may be performed with medical device **114** coupled to insertion device **112**. As shown, medical device **114** includes movable body **140** that is movable relative to a main handle body **126A**. Moveable body **140** may include a ring, for example, to receive a user's finger. In this aspect, the user may grasp handle extension **166** with one hand, and also manipulate movable body **140** with a finger (e.g., a forefinger) of the same one hand. Moveable body **140** may

be distally extended, for example, in the direction of arrow F, and distal end **130** may move from a first configuration (e.g., a bent configuration) to a second configuration (e.g., an extended configuration), for example, in the direction of arrow G. Similarly, movable body **140** may be proximally retracted, and distal end **130** of medical device shaft **128** may move from the second configuration to the first configuration. Additionally, there may be one or more intermediate configurations based on the intermediate positions of movable body **140**.

(57) As discussed above with respect to main handle body **26A** and socket body **26B**, joystick **164** may be movable relative to main handle body **126A** to further manipulate distal end **130** of medical device shaft **128**. For example, joystick **164** and main handle body **126A** may form a ball and socket assembly with a plurality of steering wires, as discussed above. Additionally, joystick **164** may be coupled to a proximal portion of main handle body **126A**.

(58) FIGS. **8A** and **8B** illustrate portions of medical device handle **126**, with main handle body **126A** as being transparent. As shown, joystick **164** may be coupled to a ball portion **134**, and ball portion **134** may be received within a cavity **144** in a proximal portion of main handle body **126A**. In this aspect, a distal portion of joystick **164** may be integrally formed with a proximal portion of ball portion **134**. Alternatively, the distal portion of joystick **164** may be coupled to the proximal end of ball portion **134**, for example, via an adhesive, one or more screws, a snap fit, or other appropriate coupling. Accordingly, movement of joystick **164** also moves ball portion **134**, for example, within cavity **144**. A plurality of steering wires **150** may be coupled to portions of ball portion **134**, and steering wires **150** may be coupled to one or more distal portions of medical device shaft **128**, as discussed above, in order for manipulation of joystick **164** to control the deflection or movement of a distal portion of medical device shaft **128**.

(59) Moreover, movable body **140** may be coupled to an actuation wire **152**, such that movement of movable body **140** relative to main handle body **126A** controls the position of distal end **130** of medical device shaft **128**. Although not shown, main handle body **126A** may include a slot in which movable body **140** may move proximally and/or distally. Steering wires **150** and actuation wire **152** may extend through medical device shaft **128**, for example, through one or more lumens, as discussed above. Additionally, steering wires **150** may be routed around the movable body **140**. Furthermore, as shown in FIG. **8A**, main handle body **126A** may be formed, for example, of two halves, and main handle body **126A** may include one or more coupling holes **144C**, and the halves may be coupled around ball portion **134** and other components, for example, via one or more screws, bolts, etc. positioned within coupling holes **144C**.

(60) Main handle body **126A** may include a cylindrical portion **154** with a channel **154A** to receive a portion of medical device shaft **128**. Although referred to as a cylindrical portion herein, it is contemplated that portion **154** may have any suitable cross-sectional shape and may be, for example, tapered. Additionally, channel **154A** may include a widened channel portion or slot **154B**. In some aspects, channel **154A** may be generally cylindrical, and slot **154B** may be generally rectangular, although other shapes are contemplated. As shown in FIG. **8B**, a crimp **154C** may be positioned around a portion of medical device shaft **128** within slot **154B**, for example, to help couple medical device shaft **128** to medical device handle **126**. Furthermore, although not shown, one or more additional wires or conduits may couple hub **136B** to actuation wire **152** (e.g., passing through handle extension **166** and a portion of main handle body **126A**), for example, to deliver energy to actuation wire **152**. As shown in FIG. **8B**, a handle cap **154D** may be coupled to a distal portion of cylindrical portion **154** to help couple portions of main handle body **126A** and to help couple medical device shaft **128** to main handle body **126A**.

(61) FIG. **9** illustrates the movement of medical device **114**. Specifically, FIG. **9** illustrates the movement of joystick **164** relative to main handle body **126A** and the movement of distal end **130** of medical device shaft **128**. Joystick **164** may be positioned in various positions, for example, **164A**, **164B**, **164C**, and **164D**, or other intermediate positions. Joystick **164** may be manipulated by one or more of the user's fingers, for example, by the user's thumb, by the user's thumb and

forefinger, etc. In one aspect, the user may grasp handle extension **166** with one hand and manipulate joystick **164** with a thumb of the same one hand. As a result, distal end **130** may be deflected to various positions, for example, **130A**, **130B**, **130C**, **130D**, or various intermediate positions, which correspond to the various joystick positions. Additionally, it is noted that joystick **164** may move in any direction, indicated by arrows H, which may deflect distal end **130** in any direction, indicated by arrows I. Furthermore, joystick **164** may move incrementally and to different extents in the various directions, for example, to deflect distal end **130** in a semi-spherical range, for example, relative to a distal end of insertion device **112**. In some aspects, depending on the deflectability of distal end **130**, movement of joystick **164** may deflect distal end **130** in a range that is larger than a semi-spherical range. Although not shown in FIG. **9**, movement or deflection of insertion device shaft **120** and/or of movable body **140** may also control the movement of distal end **130**, as discussed above. Furthermore, medical device **114** may be moved proximally or distally relative to insertion device **112** (e.g., relative to port **122**) to control the position of distal end **130** relative to the distal end of insertion device shaft **120**.

(62) FIGS. **10A-10D** illustrate various features that may be incorporated in medical device shaft **128**. As shown in FIG. **10A**, medical device shaft **128** may include a plurality of sections. For example, medical device shaft **128** may include a main portion **128A**. Main portion **128A** may span a majority of the length of medical device shaft **28**, for example, approximately 70%, approximately 75%, approximately 80%, approximately 85%, approximately 90%, approximately 95%, or more of the length of medical device shaft **128**. Additionally, medical device shaft **128** may include a distal shaft portion **128B** and a distal cautery shaft portion **128C**. Distal shaft portion **128B** may be positioned between main portion **128A** and distal cautery shaft portion **128C**. Moreover, distal cautery shaft portion **128C** may include one or more support portions **128D**, which may help to support cautery wire **168**, for example, as cautery wire **168** extends from distal cautery shaft portion **128C**. As mentioned, cautery wire **168** may extend from openings in distal cautery shaft portion **128C**. Furthermore, as discussed above, the different portions of medical device shaft **128** may have different properties, layers of material, etc., which may affect the flexibility or rigidity of the various portions.

(63) FIG. **10B** illustrates a perspective view of a lateral cross-section of main portion **128A** of medical device shaft **128**, indicated by the line **10B-10B** in FIG. **10A**. As shown in FIG. **10B**, main portion **128A** of medical device shaft **128** includes an interior multi-lumen element **156**, which may be formed of polytetrafluoroethylene (“PTFE”) or other appropriate material, for example, via extrusion or other appropriate formation process. Multi-lumen element **156** may include a durometer between approximately 20 to 60, for example, between approximately 33 and 50. Steering wires **150** may be movably positioned within respective lumens **156A** in multi-lumen element **156**, for example, spaced approximately 90 degrees apart in the cross-section of multi-lumen element **156**. Additionally, multi-lumen element **156** may include a central lumen **156B**, for example, centrally located in multi-lumen element **156**, and actuation wire **152** may be moveably positioned within central lumen **156B**.

(64) Main portion **128A** of medical device shaft **128** may include one or more outer sheaths **158A**, for example, formed of a polyether block amide (“PEBA”) or other appropriate material. In this aspect, a braiding **158C** and one or more sheaths **158D** (e.g., formed of PEBA with a durometer of approximately 60 to 80, for example, approximately 72) may be positioned between multi-lumen element **156** and outer sheath **158A**. Outer sheath **158A** may be formed via an extrusion, and may be coupled to the other components of main portion **128A** via heat shrinking. As shown, outer sheath **158A** may include a C-shaped opening **172**, which may receive a C-shaped tube **174**. Tube **174** may receive a portion of guide wire **170** (FIGS. **6A** and **6B**), for example, in order for medical device shaft **128** to be delivered to a treatment site over a guidewire.

(65) FIG. **10C** illustrates a perspective view of a lateral cross-section of distal shaft portion **128B**, indicated by the line **10C-10C** in FIG. **10A**. As shown in FIG. **10C**, distal shaft portion **128B** may

include multi-lumen element **156**, which includes respective lumens for wires **150** and actuation wire **152**, as discussed above. Distal shaft portion **128B** of medical device shaft **128** may include one or more outer sheaths **158E**, for example, formed of a polyether block amide (“PEBA”) or other appropriate material. In this aspect, a braiding **158F** and one or more intermediate sheaths (not shown) may be positioned between multi-lumen element **156** and outer sheath **158E**. Braiding **158F** and the intermediate sheath(s) may include a durometer between approximately 20 to 60, for example, between approximately 33 to 50. Outer sheath **158E** may be formed via an extrusion, and may be coupled to the other components of distal shaft portion **128B** via, e.g., heat shrinking. As shown, outer sheath **158E** may include an opening **176**, which may include a circular cross-section to form a cylindrical opening. Opening **176** may at least partially align with C-shaped tube **174**. In this aspect, a guide wire may pass through both distal shaft portion **128B** and main portion **128A**, with the guide wire being enclosed within a portion of outer sheath **158E** over distal shaft portion **128B**. For example, a guide wire may be used to deliver medical device shaft **128**, and, optionally, insertion device shaft **120**, to the treatment site. Furthermore, one or more guide wires may be delivered through and/or removed from medical device shaft **128** in order to treat the treatment site. (66) FIG. **10D** illustrates distal end **130** of medical device shaft **128**, including distal cautery shaft portion **128C** extending from distal shaft portion **128B**. As mentioned, cautery wire **168** may extend from portions of distal cautery shaft portion **128C**. Additionally, distal cautery shaft portion **128C** may include a distal opening **178**. Although not shown, opening **176** in distal shaft portion **128B** may transition from an exterior of medical device shaft **128** to a central portion of medical device shaft **128** over a portion distal cautery shaft portion **128C**. In this aspect, a guide wire (not shown) may pass through distal opening **178** in distal cautery shaft portion **128C** and into opening **176** in distal shaft portion **128B** (FIG. **10C**). Furthermore, although not shown, portions of steering wires **150** may be coupled to portions of distal cautery shaft portion **128C**, for example, such that movement of joystick **164** moves distal cautery shaft portion **128C**, as discussed above, for example, with respect to FIG. **9**.

(67) FIGS. **11A**, **11B**, and **12A-12C** illustrate various features of guide wire device **162**. Guide wire device **162** includes a handle portion **162A** and guide wire **170**. Handle portion **162A** may include a casing **162B**, for example, for a user to hold. Handle portion **162A** also includes a button **162C** and a roller **162D**. Button **162C** may be positioned within a channel **162E** in roller **162D**. As shown in FIG. **11B**, button **162C** may be coupled to a portion of guide wire **170**, for example, to a pull wire **170A**. In one aspect, a sleeve **162F** may be crimped to pull wire **170A**, and sleeve **162F** may be coupled to button **162C**, for example, via an adhesive, one or more screws, a snap fit, or other appropriate coupling. In this aspect, movement of button **162C**, for example, within channel **162E**, may control the movement, and thus actuation, of pull wire **170A**. Additionally, roller **162D** may be rotatably coupled to casing **162B**, for example, via overlapping extensions, as shown in FIG. **11B**. In this aspect, rotating roller **162D** may also rotate button **162C**, and thus also rotate pull wire **170A**. Although not shown, guide wire device **162** may include one or more frictional or locking elements, for example, to limit and/or lock the movement of button **162C** and/or roller **162D**.

(68) Guide wire **170** may include a main wire portion **170B** and a distal wire portion **170C**. Main wire portion **170B** may be formed of a hollow wire, for example, formed of nitinol or other appropriate material, and may be fixedly coupled to casing **162B**. Distal wire portion **170C** may be formed of a flexible material, for example, PEBA, rubber, PTFE, etc. Furthermore, pull wire **170A** may be a solid wire, for example, formed of nitinol, stainless steel, or other appropriate material. Distal wire portion **170C** may articulate, for example, based on the position of pull wire **170A**. For example, a distal end of pull wire **170A** may be coupled to a distal portion of distal wire portion **170C**, for example via an adhesive, heat shrinking, crimping, or other appropriate coupling.

(69) FIG. **11B** shows distal wire portion **170C** as being partially transparent. Distal wire portion **170C** may include a closed distal end. In this aspect, a coupling **170D** between a portion of pull wire **170A** and distal wire portion **170C** may only be formed over a circumferential portion of pull

wire **170A**. In this aspect, less than an entire circumference of the portion of pull wire **170A** that is positioned within distal wire portion **170C** may form coupling **170D**. For example, a semi-circular or quarter-circular portion of the outer circumference of pull wire **170A** may be coupled to an interior of distal wire portion **170C** at coupling **170D**. In this aspect, movement of pull wire **170A**, for example, via button **162C** and/or roller **162D**, may articulate distal wire portion **170C**. Furthermore, if distal wire portion **170C** is positioned within a portion of medical device shaft **128**, for example, distal cautery shaft portion **128C** and/or distal shaft portion **128B**, then movement of distal wire portion **170C** may also articulate the portion of medical device shaft **128**.

(70) FIGS. **12A-12C** illustrate various movements of handle portion **162A** and guide wire **170**. For example, FIG. **12A** illustrates button **162C** being actuated distally (i.e., in the direction of arrow J) within channel **162E** in roller **162D**. In this aspect, distal articulation of button **162C** articulates pull wire **170A** distally. Based on the connection of pull wire **170A** to distal wire portion **170C**, the distal articulation of pull wire **170A** articulates distal wire portion **170C** in a first direction, for example, upward (i.e., in the direction of arrow K).

(71) FIG. **12B** illustrates button **162C** being actuated proximally (i.e., in the direction of arrow L) within channel **162E** in roller **162D**. In this aspect, proximal articulation of button **162C** articulates pull wire **170A** proximally. Based on the connection of pull wire **170A** to distal wire portion **170C**, the proximal articulation of pull wire **170A** articulates distal wire portion **170C** in a second direction, for example, downward (i.e., in the direction of arrow M).

(72) Furthermore, FIG. **12C** illustrates roller **162D**, and thus button **162C**, being rotated clockwise or counterclockwise (i.e., in the direction of arrows N) relative to casing **162B**. In this aspect, rotation of button **162C** rotates pull wire **170A**. Furthermore, button **162C** may be articulated, for example, either proximally or distally, as discussed above, and pull wire **170A** and distal wire portion **170C** may be rotated and articulated in a plurality of directions (i.e., in the direction of arrows O). In this aspects, and as discussed above, guide wire **170** may help to guide and/or articulate a portion of medical device shaft **128**.

(73) FIG. **13** illustrates another exemplary medical system **210**, according to aspects of this disclosure. System **210** includes an insertion device **212**, for example, similar to insertion device **12**. Medical system **210** also includes a medical device **214**. Medical system **210** may include an adapter **216**, for example, coupling medical device **214** to a port **222** of insertion device **212**. Furthermore, medical system **210** may include a guide wire device **262**, for example, similar to guide wire device **162**, and a cautery wire **268**, for example, similar to cautery wire **168**. Medical device **214** may be coupled to and/or delivered to a treatment site through a handle **218** and shaft **220** of insertion device **212**, as discussed above. Guide wire device **262** may include a guide wire **270**, which may be coupled to medical device **214**, for example, via a slot or opening in a portion of a medical device shaft **228** or medical device handle **226**. Additionally, adapter **216** may include an adapter slot or opening **216D**. In this aspect, a portion of guide wire **270** may pass through opening **216D**. For example, the portion of guide wire **270** may move within opening **216D** as medical device shaft **228** moves (e.g., proximally or distally) relative to adapter **216** and port **222** of insertion device **212**.

(74) Medical device **214** includes a medical device handle **226** and medical device shaft **228**. Medical device handle **226** may include a movable handle portion or a main handle body **226A** and a stationary handle portion or a joint (e.g., a ball joint), referred to herein as socket body **226B**, with the main handle body **226A** being movable relative to and positioned proximal of socket body **226B**. As discussed with respect to medical device **14**, socket body **226B** may abut a portion of adapter **216**, and, although not shown, main handle body **226A** may be coupled to a plurality of steering wires and an articulation wire. The articulation wire may be coupled to movable body **240** on main handle body **226A**. Furthermore, main handle body **226A** and socket body **226B** may form a ball and socket joint. In this aspect, movement of main handle body **226A** relative to socket body **226B** controls the movement of one or more steering wires and controls a deflection of distal end

230 of medical device shaft **228**, for example, relative to the distal end of insertion device shaft **220**. The position of medical device handle **226** relative to port **222** of insertion device handle **218** may also control the position of distal end **230** relative to the distal end of insertion device shaft **220**. Furthermore, as discussed above, main handle body **226A** may include one or more ports (not shown) or hubs **236B**, for example, for coupling of fluid, suction, or energy sources to medical device **214**. In this aspect, a user may hold and/or manipulate handle **218** of insertion device **212** with one hand, and may hold and/or manipulate main handle body **226A** (e.g., relative to socket body **226B**) with another hand.

(75) Moreover, as discussed above with respect to FIGS. **10A-10D**, **11A**, **11B**, and **12A-12C**, guide wire device **262** may help to manipulate distal end **230** of medical device shaft **228**. For example, guide wire **270** may extend through a portion of medical device shaft **228**, for example, to distal end **230**. As discussed above, guide wire device **262** includes a handle portion **262A** with a casing **262B**. A button **262C** and a roller **262D** may be movable relative to casing **262B** of guide wire device **262**, for example, may be articulated proximally, distally, or rotated, to control the movement of a pull wire (not shown) in guide wire **270**. The movement of the pull wire, and thus the movement of guide wire **270**, may help to guide and/or articulate a distal portion of medical device shaft **128**, for example, in a plurality of directions.

(76) Various aspects discussed herein may allow for an insertion device and a medical device (e.g., medical device **12**) to be delivered to a treatment site, for example, to perform endoscopic submucosal dissection (“ESD”) or otherwise treat the treatment site. One or more portions of the medical device (e.g., electrode **38**) may be selectively positioned relative to the insertion device by movement of steering wires **50**. Moreover, the selective manipulation of one or more wires **50**, for example, via movement of main handle body **26A** relative to socket body **26B** may allow for medical device **14** to approach the treatment site at one or more angles different than insertion device **12**, which may help the user to perform the treatment at the treatment site. Specifically, the manipulation of one or more wires **50** may allow for medical device shaft **28** to be deflected (e.g., pivoted or reoriented) up to approximately 90 degrees in each direction relative to distal end **20A** of insertion device shaft **20**.

(77) In addition to the different angles of approach, medical systems **10**, **110**, and **210** may provide the user with increased control of the delivered medical device and/or to treat and/or view the treatment site. Wires **50**, **150** may be controlled by a medical device handle **26**, **126**, **226**, allowing the user a convenient mechanism to control the position of the medical device shaft relative to the insertion device shaft. Additionally, the force applied and direction of movement (e.g., on main handle body **26A**, joystick **164**, or main handle body **226A**) to the wires may impart different forces and directions of movement to control the amount of pivoting or reorientation of the medical device shaft. The medical systems, including the wires, may allow for the distal portions of the medical device shafts to be pivoted or reoriented to any position or direction within the semispherical range (or greater) extending from the distal end of the insertion device. In some aspects, for example, medical systems **10**, **110**, and **210** may allow for the distal portion of the medical device shaft to pivot or reorient separate from the insertion device. Furthermore, the medical device handle may be moved longitudinally, e.g., proximally or distally, relative to the insertion device handle to control the longitudinal position of the distal end of the medical device shaft relative to the distal end of the insertion device shaft. An end effector (e.g., electrode **38**) positioned at distal end **30** of medical device shaft **28** may also be separately movable or activated, for example, via movable body **40**.

(78) As discussed, the medical systems discussed herein may allow for a single operator to control various aspects of the medical systems. For example, referring to FIG. **1**, a user may hold and/or control insertion device **12** with a first hand. The user may manipulate control device **24**, for example, to deflect distal end **20A** of insertion device shaft **20**. The user may also hold and/or control medical device **14** with a second hand. The user may manipulate the second hand (e.g., via

wrist movement) to control manipulate main handle body **26A** relative to socket body **26B**, for example, to manipulate distal end **30** of medical device shaft **28**. Furthermore, the user may manipulate a finger (e.g., the thumb on the second hand) to manipulate movable body **40**, for example, to extend or retract or otherwise actuate an end effector (e.g., electrode **38**) relative to distal end **30** of medical device shaft **28** via actuation wire **52**. In these aspects, a single user may control various movements and/or functions of medical systems **10**, **110**, **210**. Moreover, the various movements, for example, movement of main handle body **26A** relative to socket body **26B**, may be intuitive and/or easy for the user to learn and execute during a procedure to treat the treatment site.

(79) Steering wires **50** may be flexible and may be formed of metal (e.g., stainless steel), plastic, or include a shape memory metal (such as nitinol), a shape memory polymer, a polymer, or any combination of biocompatible materials, or another appropriate biocompatible material. Steering wires **50** may be relatively thin compared to medical devices **14**, **114**, **214**. In this aspect, steering wires **50** may not significantly increase the cross-sectional size of medical devices **14**, **114**, **214**, may not otherwise interfere with the movement of medical devices **14**, **114**, **214**. Similarly, actuation wire **52** may be formed of a metal (e.g., stainless steel), plastic, or include a shape memory metal (such as nitinol), a shape memory polymer, a polymer, or any combination of biocompatible materials, or another appropriate biocompatible material. As discussed above, actuation wire **52** may be conductive, for example, to deliver electrical energy to electrode **38**, cautery wire **168**, etc. As discussed herein, other components of medical systems **10**, **110**, **210** may be formed of a plastic, polymer, any combination of biocompatible materials, or another appropriate biocompatible material. In some aspects, various components or portions of components may be formed via an extrusion process, injection molding, additive manufacturing, etc.

(80) Accordingly, various aspects discussed herein may help to improve the efficacy of treatment and/or recovery from a procedure, for example, a procedure to treat a treatment side. Various aspects discussed herein may help to reduce and/or minimize the duration of the procedure, and/or may help reduce risks of inadvertent contact with tissue or other material during delivery, repositioning, or removal of a medical system for the procedure. Additionally, various aspects discussed herein (e.g., medical device **114** and guide wire device **162**) may be packaged as a kit to be used to treat a patient, for example, by coupling the kit to an insertion device.

(81) While principles of this disclosure are described herein with reference to illustrative aspects for various applications, it should be understood that the disclosure is not limited thereto. Those having ordinary skill in the art and access to the teachings provided herein will recognize additional modifications, applications, aspects, and substitution of equivalents all fall within the scope of the aspects described herein. Accordingly, the disclosure is not to be considered as limited by the foregoing description.

Claims

1. A medical device system, comprising an insertion device, including: an insertion device handle, including a port on a handle body, wherein the port includes an extension extending radially outward; and an insertion device shaft extending from the insertion device handle, wherein the insertion device shaft includes a working channel connected to the port; and a medical device, including: a medical device handle, including a movable handle portion and a stationary handle portion, wherein the movable handle portion includes a ball portion movably positioned within a cavity in the stationary handle portion; an adapter positioned between the port on the insertion device handle of the insertion device and the medical device handle, wherein the adapter includes a projection extending radially inward, wherein the extension is proximal to the projection; and a medical device shaft, wherein the medical device shaft is configured to be delivered through the

port in the insertion device handle and through the working channel in the insertion device shaft, wherein movement of the movable handle portion relative to the stationary handle portion controls movement of a distal portion of the medical device shaft.

2. The medical device system of claim 1, further comprising a plurality of steering wires, wherein each of the plurality of steering wires is coupled to the ball portion at a proximal end and to the distal portion of the medical device shaft at a distal end.
3. The medical device system of claim 2, wherein the distal ends of the plurality of steering wires are coupled to a ring at the distal portion of the medical device shaft.
4. The medical device system of claim 3, wherein the plurality of steering wires includes four steering wires coupled to the ball portion, wherein the ball portion includes four wire mounts at locations 90 degrees apart from each other around a circumference of the ball portion, and wherein each wire mount includes a crimping slot to couple each steering wire to the ball portion.
5. The medical device system of claim 1, wherein the medical device includes an actuation wire and a movable body coupled to the actuation wire, wherein the actuation wire is movable relative to the medical device shaft by movement of the movable body relative to the stationary handle portion.
6. The medical device system of claim 5, wherein the medical device includes a distal electrode, and wherein movement of the movable body manipulates the actuation wire to extend or retract the distal electrode relative to a distal end of the medical device shaft.
7. The medical device system of claim 1, wherein the medical device handle includes a fluid port or a cautery hub.
8. The medical device system of claim 1, further comprising a cautery hub on the medical device handle and an actuation wire extending through at least a portion of the medical device handle and the medical device shaft, wherein the actuation wire is electrically connected to the cautery hub.
9. The medical device system of claim 1, wherein the adapter includes an arced portion or is at least partially flexible.
10. The medical device system of claim 9, wherein the adapter is removably coupled to the insertion device and includes one or more slits on a distal portion, and wherein the adapter includes a lock nut that is movable along the distal portion to control a width of each of the one or more slits.
11. The medical device system of claim 1, further comprising a guide wire device, wherein the guide wire device includes a guide wire handle and a guide wire.
12. The medical device system of claim 11, wherein the guide wire includes a main wire portion, a distal wire portion, and a movable pull wire coupled to the distal wire portion, wherein the guide wire handle includes a casing, a roller rotatably coupled to the casing, and a button that is movable within a channel in the roller, and wherein the roller is coupled to the pull wire to control extension or rotation of the pull wire.
13. The medical device system of claim 12, wherein a portion of the pull wire is coupled to the distal wire portion via a coupling, and wherein the coupling extends over less than an entire outer circumference of the portion of the pull wire.
14. The medical device system of claim 1, wherein the movable handle portion is a joystick, and wherein the stationary handle body is a main handle body positioned distal to the joystick.
15. A medical system, comprising: a medical device, including: a medical device handle, including a movable handle portion and a stationary handle portion, wherein the movable handle portion includes a ball portion movably positioned within a cavity in the stationary handle portion; a medical device shaft extending from the medical device handle; and a plurality of wires, wherein the plurality of wires are coupled to the ball portion of the movable handle portion and to a ring at a distal portion of the medical device shaft, wherein movement of the ball portion of the movable handle portion within the cavity in the stationary handle portion manipulates a distal portion of the medical device shaft; and a guide wire device, including: a guide wire handle; and a guide wire,

wherein the medical device shaft includes one or more openings or tubes along one or more outer portions of the medical device shaft configured to receive a portion of the guide wire, wherein the guide wire device includes a main wire portion, a distal wire portion, and a movable pull wire coupled to the distal wire portion, wherein the guide wire handle includes a casing, a roller rotatably coupled to the casing, and a button that is movable within a channel in the roller, and wherein the roller is coupled to the pull wire to control extension or rotation of the pull wire.

16. The medical system of claim 15, further comprising: an insertion device, wherein the insertion device includes: an insertion device handle having a port to receive a portion of the medical device shaft and a portion of the guide wire; an insertion device shaft extending from the insertion device handle, wherein the insertion device shaft includes a lumen in communication with the port; and a control device coupled to a portion of the insertion device handle, wherein the control device is movable to control a position of a distal end of the insertion device shaft.

17. A medical device, comprising: a medical device handle, including a joystick portion and a stationary handle portion, wherein the joystick portion is positioned distal to the stationary handle portion and includes a ball portion movably positioned within a cavity in the stationary handle portion; a medical device shaft extending from the medical device handle; and a plurality of wires, wherein the plurality of wires are coupled to the ball portion of the joystick portion and to a ring at a distal portion of the medical device shaft, wherein movement of the ball portion of the movable handle portion within the cavity in the stationary handle portion manipulates a distal portion of the medical device shaft wherein the ball portion includes: a plurality of wire mounts disposed around a circumference of the ball portion, wherein each wire mount includes (i) a crimping slot, (ii) a wire slot, and (iii) a wire opening disposed between the crimping slot and the wire slot.

18. The medical device of claim 17, further comprising: a movable body coupled to the stationary handle portion; and an actuation wire extending from the movable body to an end effector at a distal end of the medical device shaft, wherein movement of the movable body controls an extension or retraction of an end effector.

19. The medical device of claim 18, further comprising a guide wire device, wherein the guide wire device includes a guide wire handle and a guide wire.
